METHOD FOR DYNAMIC COMPOSITION OF BARE-METAL COMPUTING RESOURCES

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Title: METHOD FOR DYNAMIC COMPOSITION OF BARE-METAL COMPUTING RESOURCES

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Abstract

A method for dynamically composing and provisioning a logical, bare-metal computer from physically disaggregated pools of hardware resources. The method comprises: a control plane receiving a request specifying a quantity of computer resources (e.g., number of CPUs, amount of memory); identifying available, disaggregated resources from their respective pools; transmitting a command to a data fabric switch to establish a set of persistent optical links interconnecting the identified resources; and exposing the resulting collection of interconnected resources as a single, unified, bare-metal computer to a user or task. Upon task completion, the links are dissolved, and the resources are returned to the pools.

Background of the Invention

Cloud computing and virtualization provide resource flexibility by creating virtual machines, which are software-based slices of a single, monolithic physical server. While this allows for

rapid provisioning, it introduces performance overhead from the hypervisor layer. Provisioning a true "bare-metal" server, by contrast, is a slow, manual process involving physical hardware allocation. There is a need for a method that combines the speed and flexibility of cloud provisioning with the performance of bare-metal hardware.

Summary of the Invention

The present invention is a method for the dynamic composition of bare-metal computing resources, which leverages a physically disaggregated hardware architecture and an optical data fabric switch. This method provides "bare-metal-as-a-service," where a user can request a specific hardware configuration and have it provisioned on-demand.

The process is managed by a software control plane. It begins when a user or automated orchestrator sends a request to the control plane for a computer with specific attributes (e.g., 8 CPU cores, 1TB of RAM, 2 GPUs). The control plane consults an inventory of available hardware in the disaggregated resource pools (CPU pool, memory pool, GPU pool) and selects the necessary components. It then sends a series of commands to an optical data fabric switch, instructing it to create the specific point-to-point optical links that would connect these selected components as if they were on a single motherboard. The resulting logical computer is then provisioned to the user. When the user's task is complete, the control plane instructs the switch to dissolve the links, and the individual hardware components are returned to their respective pools, available for the next composition request.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the steps of the dynamic composition and decomposition method.

Detailed Description of the Invention

Referring to **FIG. 1**, the method is shown as a sequence of steps. The process begins with a control plane receiving a request for a bare-metal computer with specified resources (101). The control plane identifies and reserves the required available components from the disaggregated CPU, memory, and other resource pools (102). It then sends a command to the optical data fabric switch (103), instructing it to establish the necessary optical links to interconnect the reserved components. The switch executes the command, creating a logical computer (104). This newly composed computer is then provisioned to the user for a finite duration (105). Once the task is complete and the user releases the machine, the control plane sends a teardown command to the switch (106), which dissolves the optical links. Finally, the individual hardware components are returned to the available resource pools (107).

Claims

What is claimed is:

1. A method for dynamically composing a computer, the method comprising the steps of:

    - receiving, at a software control plane, a request for a logical computer specifying a set of hardware resources;
    - identifying available, physically disaggregated hardware components from a plurality of resource pools that satisfy the request;
    - instructing a data fabric switch to establish a plurality of communication links that interconnect the identified hardware components; and
    - provisioning the interconnected hardware components as a single logical computer.

2. The method of claim 1, further comprising the steps of:

    - receiving a signal indicating the logical computer is no longer needed;
    - instructing the data fabric switch to dissolve the communication links; and

- returning the hardware components to the plurality of resource pools.

3. The method of claim 1, wherein the logical computer is a bare-metal computer, without a hypervisor layer between the hardware and an operating system.

4. The method of claim 1, wherein the resource pools include a pool of processors and a pool of memory modules housed in separate chassis.

5. The method of claim 1, wherein the communication links are massively parallel optical links.

6. The method of claim 5, wherein the data fabric switch is an optical data fabric switch.

Drawings

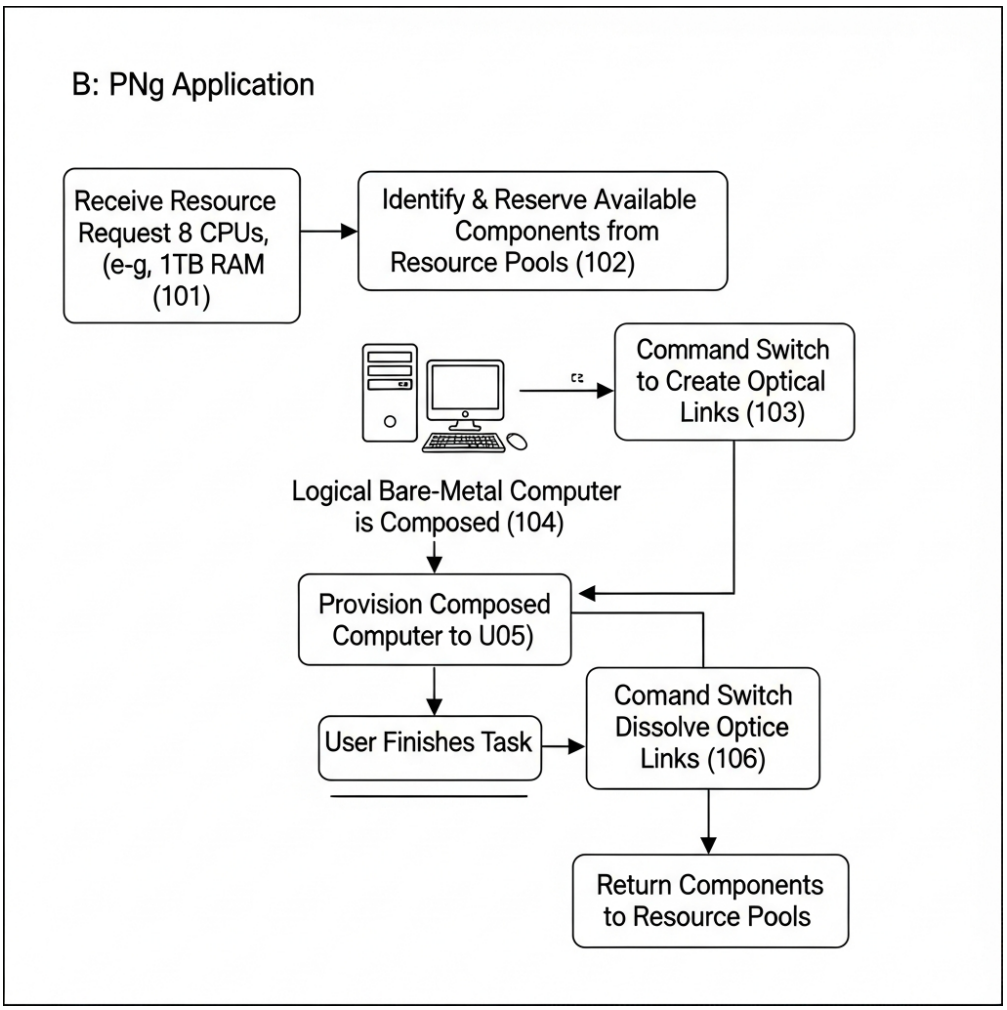


Figure 1: Flowchart illustrating the steps of the dynamic composition and decomposition method.